

Tingwo Security Company case and fundamental database principles.

1. Tingwo Security Company Analysis

- **Additional Information to Store:** To ensure data integrity, the system should include supplier contact details, item unit costs, reorder levels, staff hire dates, and historical stock adjustment logs.
- **Key Personnel:**
 - **Store Manager:** Oversees inventory, suppliers, and approvals.
 - **Store Clerk:** Manages daily requisitions and records issuance.
 - **General Manager:** Reviews weekly stock and activity reports.
- **Identified Relations (Tables):**
 1. **Staff:** Employee details for guards, supervisors, and managers.
 2. **Zones:** Geographical service areas and assignments.
 3. **Inventory:** Master list of all security equipment.
 4. **Issuance:** Transaction records of items given to personnel.
- **Sample Table Structures:**
 - **Staff Table:** Staff_ID (PK), Name, Role, Zone_ID (FK).
 - **Inventory Table:** Item_Code (PK), Category, Stock_Level, Last_Restock_Date.

2. Practical Scenario:

Library Management System

- **Key Data Sets (Entities):** Includes **Books** (catalog), **Members** (borrowers), and **Loans** (transactions).
- **Relationships:**
 - **One-to-Many:** One member can have multiple loan records.
 - **One-to-Many:** One unique book can be loaned out to different members over time.
- **User Access Levels:**
 - **Librarian:** Full CRUD (Create, Read, Update, Delete) access for books and returns.
 - **Student:** Limited access to search catalog and view personal loan history.
 - **Administrator:** System-level control over security and user accounts.
- **Key Combinations:**
 - **Primary Keys (PK):** ISBN (Books), Student_ID (Members), and Loan_ID (Loans).
 - **Foreign Keys (FK):** The **Loans** table uses ISBN and Student_ID as foreign keys to maintain relational links.